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THE BLENDING AND TRANSMISSION OF HYDROGEN AND NATURAL GAS IN TRANSMISSION AND DISTRIBUTION PIPELINES

Arash Khabbazi[*1], Ri Li[*2], John Quinn[‡3], Mina Hoorfar[*4]
[*]: School of Engineering, University of British Columbia,
1137 Alumni Ave, Kelowna, BC V1V 1V7, Canada
[‡]: Renewable Gas Supply, FortisBC,
16705 Fraser Hwy, Surrey, BC V4N 0E8
[1]: arashjkh@student.ubc.ca
[2]: sunny.li@ubc.ca
[3]: John.Quinn@fortisbc.com
[4]: mina.hoorfar@ubc.ca

ABSTRACT

Hydrogen can be used for storing surplus electrical energy from the grid during periods of low demand for electrical energy. Hydrogen can also be used to carry the electrical energy generated from renewable energy sources such as solar and wind farms that are not connected to the grid. For these applications, electrical energy is consumed to generate hydrogen. In other words, the electrical energy is converted to hydrogen as gaseous fuel, which is referred to as the power-to-gas technology. As a result, hydrogen is used as a green energy carrier due to its zero emission.

Hydrogen generated from these power-to-gas applications has to be transported to the place of utilization. The most cost-effective transmission solution is to use the natural gas (NG) transmission system by injecting hydrogen into the natural gas pipeline, in which the two gaseous fuels are transmitted as a mixture, called H2NG. Due to different thermodynamic parameters of the two gases, H2NG has significant changes in pipeline transport conditions as compared to natural gas transmission. These changes are associated with pressure drop and temperature profiles in the pipeline. The changes also vary with the high-pressure, medium-pressure and low-pressure pipelines. Overall, H2NG has beneficial influence on the conditions of its transmission. However, the molar fraction of hydrogen in H2NG should not exceed 15%–20% due to the drop in heating value. Differences between transporting H2NG and natural gas also take place at compression and decompression stations.

One objective of this talk is to provide a brief review of the research work on the transmission of H2NG in pipeline as discussed above. This talk will also focus on the future research on the injection of hydrogen into local natural gas infrastructure or even in the natural gas supply to a single building. As compared to the transmission in main and regional pipelines, the injection into local pipeline features low pipeline pressure and is subject to dynamically changing demand, which could affect end-use NG appliances. Experimental and analytical studies are required to understand how a dynamically changing supply of H2NG can be implemented with steady energy quality in the pipeline for varied demand functions.

Keywords: hydrogen, natural gas, pipeline, transmission